

Test for Equality of Multinomial Distributions

Presentation Outline

1 Variables and Data

2 Test for Equality of Multinomial Distributions

Variables and Data Collection

- Variable 1
 - I categories
- Variable 2
 - J categories
- Data Collection and Summary
 - Stratified samples from I populations or experiment with I treatment groups.
 - Cross-classify observations according to categories of two variables
 - Form into contingency table

Example - 3 x 4 Contingency Table

Variable 1	Variable 2				Total
	Cat. 1	Cat. 2	Cat. 3	Cat. 4	
Cat. 1	Y_{11}	Y_{12}	Y_{13}	Y_{14}	n_1
Cat. 2	Y_{21}	Y_{22}	Y_{23}	Y_{24}	n_2
Cat. 3	Y_{31}	Y_{32}	Y_{33}	Y_{34}	n_3
Total	$Y_{.1}$	$Y_{.2}$	$Y_{.3}$	$Y_{.4}$	n

Null and Alternative Hypotheses

- Definitions:
 - p_{ij} = population proportion of category j of Variable 2 in category i of Variable 1.
- Null Hypothesis:
 - Distribution of Variable 2 is the same for all categories of Variable 1
 - H_0 : the set of proportions $p_{i1}, p_{i2}, \dots, p_{iJ}$ are the same for all $i = 1, 2, \dots, I$
- Alternative Hypothesis:
 - Distribution of Variable 2 is different for at least one category of Variable 1
 - H_a : At least one set of proportions $p_{i1}, p_{i2}, \dots, p_{iJ}$ are different for $i = 1, 2, \dots, I$

Expected Values under H_0

If H_0 is true, then each proportion $p_{ij} = p_{.j}$ for all $i = 1, 2, \dots, l$.

n_i = fixed number of observations in category i of Variable 1.

Expected value of each cell in contingency table under H_0 :

$$E(Y_{ij}) = n_i p_{.j}$$

Expected Values under H_0

Estimate unknown probabilities $p_{.j}$ with sample data:

$$\hat{p}_{.j} = \frac{Y_{.j}}{n}$$

Estimated expected value of each cell in the contingency table under H_0 :

$$\widehat{E}(Y_{ij}) = n_i \hat{p}_{.j} = n_i \left(\frac{Y_{.j}}{n} \right) = \frac{n_i Y_{.j}}{n} = \frac{\text{row total} * \text{column total}}{\text{table total}}$$

Test Statistic

Test statistic is based on comparing $\widehat{E}(Y_{ij})$ to the observed values Y_{ij}

$$\chi^2 = \sum_{i=1}^I \sum_{j=1}^J \frac{(Y_{ij} - \widehat{E}(Y_{ij}))^2}{\widehat{E}(Y_{ij})}$$

- Sign of difference is not important - only the magnitude.
- Differences are weighted by the estimated expected value.

P-value

Assume $\widehat{E}(Y_{ij}) \geq 5$ for all $i = 1, 2, \dots, I$ and $j = 1, 2, \dots, J$

Distribution of X^2 is well-approximated by the Chi-square distribution with $(I - 1)(J - 1)$ degrees of freedom.

P-value:

$$P\left(\chi^2_{(I-1)(J-1)} > X^2\right)$$

Example - Test of Equality

Finger Lakes Homes manufactures three models of prefabricated homes: a two-story colonial, a log cabin, and an A-frame. To help in product-line planning, management would like to study customer satisfaction with the three home styles so they conduct a survey of 120 owners of each style.

Variable 1 - Home Style

- Categories: Colonial, Cabin, A-frame

Variable 2 - Would they repurchase the same style?

- Categories: Yes, No, Maybe

Example - Test of Equality

	Repurchase?			Total
	Yes	No	Maybe	
Home Style				
Colonial	90	17	13	120
Cabin	81	20	19	120
A-frame	83	31	6	120
Total	254	68	38	360

Example - Test of Equality



Example - Test of Equality

Null Hypothesis:

- H_0 : The distribution of the Repurchase variable is the same for all Home Style customer populations.
- $H_0 : p_{ij} = p_{.j}$ for all $i = 1, 2, 3$ and $j = 1, 2, 3$

Alternative Hypothesis:

- H_a : At least one of the Home Style customer populations has a different distribution of the Repurchase variable.
- H_a : at least one $p_{ij} \neq p_{.j}$ for $i = 1, 2, 3$ and $j = 1, 2, 3$

Example - Test of Equality

Observed Data:

	Repurchase?			Total
	Yes	No	Maybe	
Home Style				
Colonial	90	17	13	120
Cabin	81	20	19	120
A-frame	83	31	6	120
Total	254	68	38	360

Example - Test of Equality

Expected Values if H_0 is true:

Home Style	Repurchase?			Total
	Yes	No	Maybe	
Colonial	84.67	22.67	12.67	120
Cabin	84.67	22.67	12.67	120
A-frame	84.67	22.67	12.67	120
Total	254	68	38	360

Example - Test of Equality

Test Statistic:

$$\begin{aligned} \chi^2 &= \frac{(90 - 84.67)^2}{84.67} + \frac{(81 - 84.67)^2}{84.67} + \dots + \frac{(6 - 12.67)^2}{12.67} \\ &= 12.006 \end{aligned}$$

Example - Test of Equality

P-value:

All estimated expected values are ≥ 5 , so we can use the $\chi^2_{(I-1)(J-1)} = \chi^2_{2*2} = \chi^2_4$ distribution to approximate the distribution of X^2 .

$$P(\chi^2_4 > 12.006) < 0.0173$$

Conclusion:

We have moderately strong evidence that the distribution of the Repurchase variable is different for at least one of the Home Styles in this population of customers.

Comparison of Chi-square Tests

- Different Null and Alternative Hypotheses
- Different Derivations of Estimated Expected Values
- Same Calculation of Estimated Expected Values
- Same Test Statistic
- Same P-value
- Different Conclusions